

Electric Vehicles

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Problem Statement:

- Electric Vehicle manufacturers struggle to compete with TESLA in terms of number of vehicles based on Washington State database; with TESLA holding 45% and the closest competitor has 8%

45%
TESLA

8%
Closest competitor

Objectives:

- Identify the variables that affect the electric vehicles market
- Visualize the growth of the EV market over the years
- Determine the percentage of CAFV-eligible vehicles
- Compare the percentages of CAFV-eligible vehicles between TESLA and the other manufacturers

Objectives:

- Determine the percentages of each battery type
- Compare the EV range between TESLA and the other manufacturers
- Provide data-driven recommendations for other manufacturers to compete with TESLA

Limitations

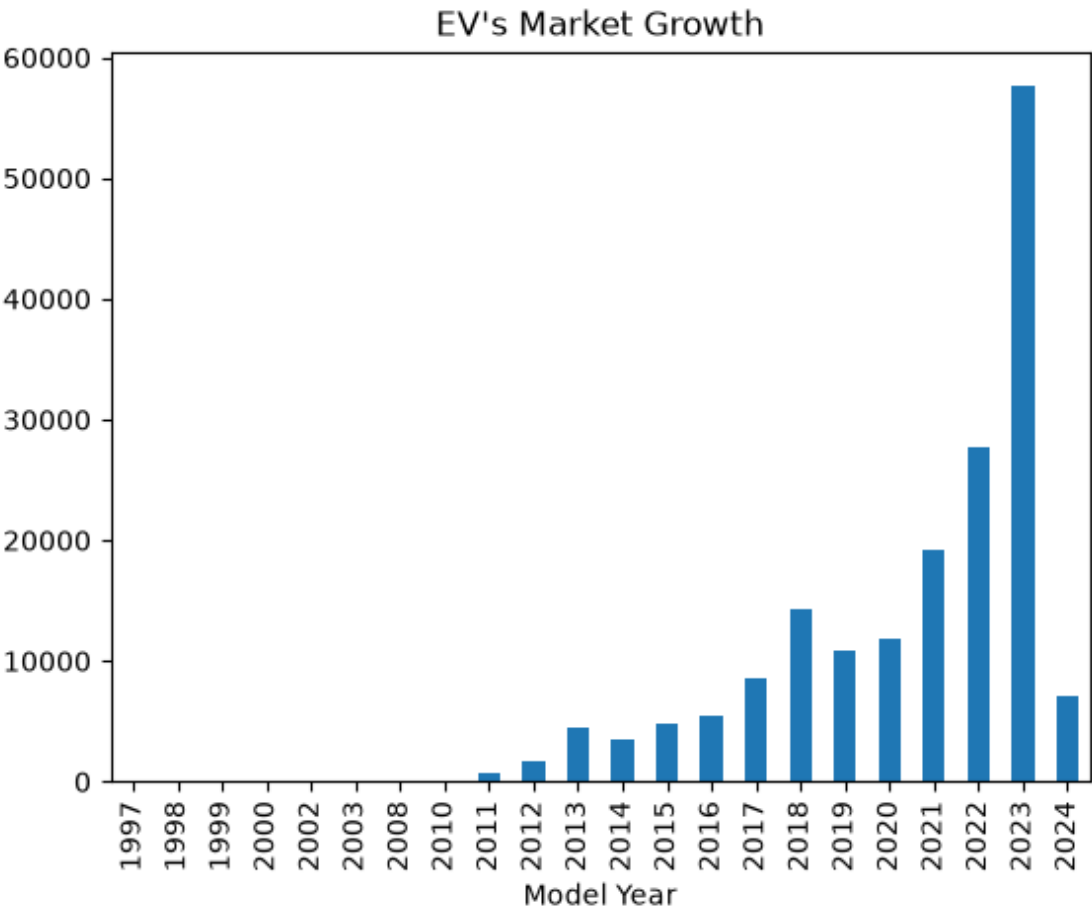
51.69%

Missing Electric Range Values

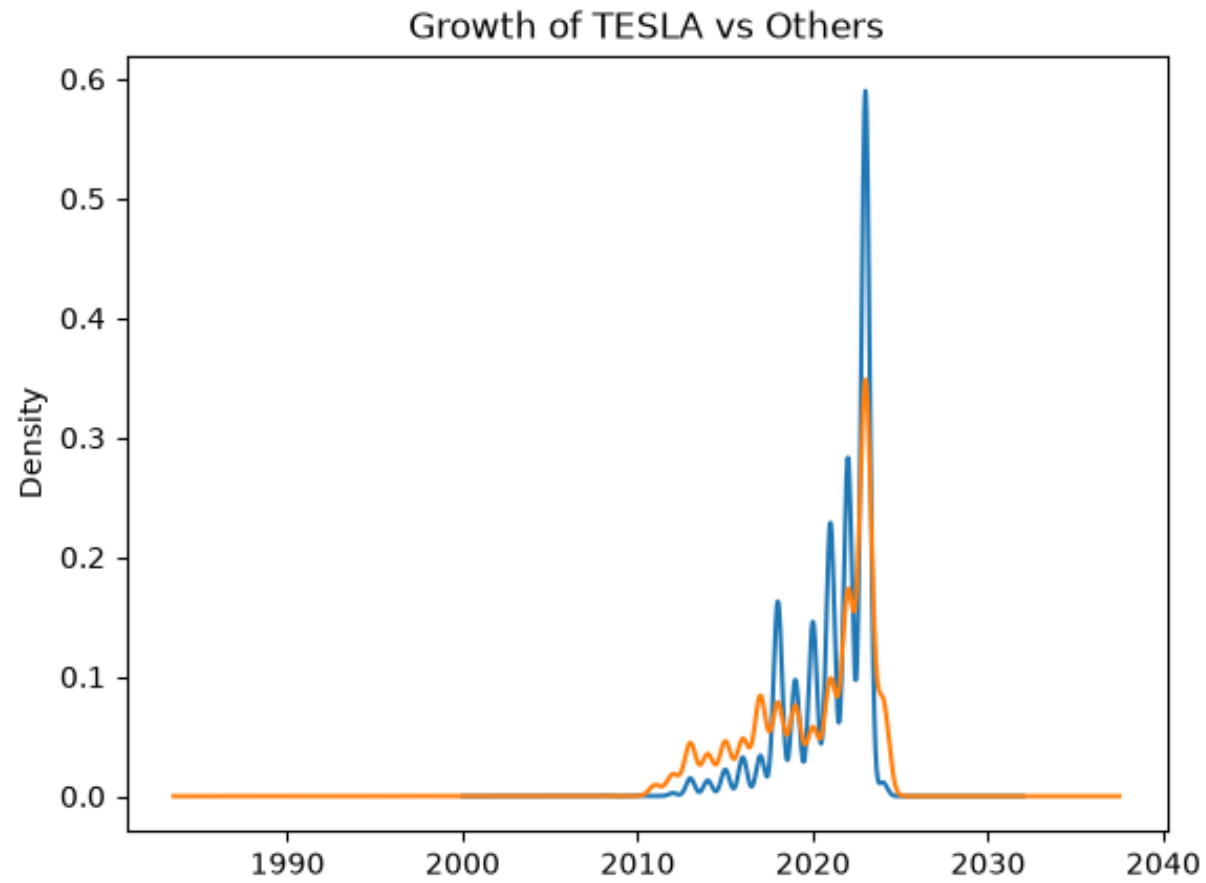
98.12%

Missing Base MSRP

Market Growth



Market Growth



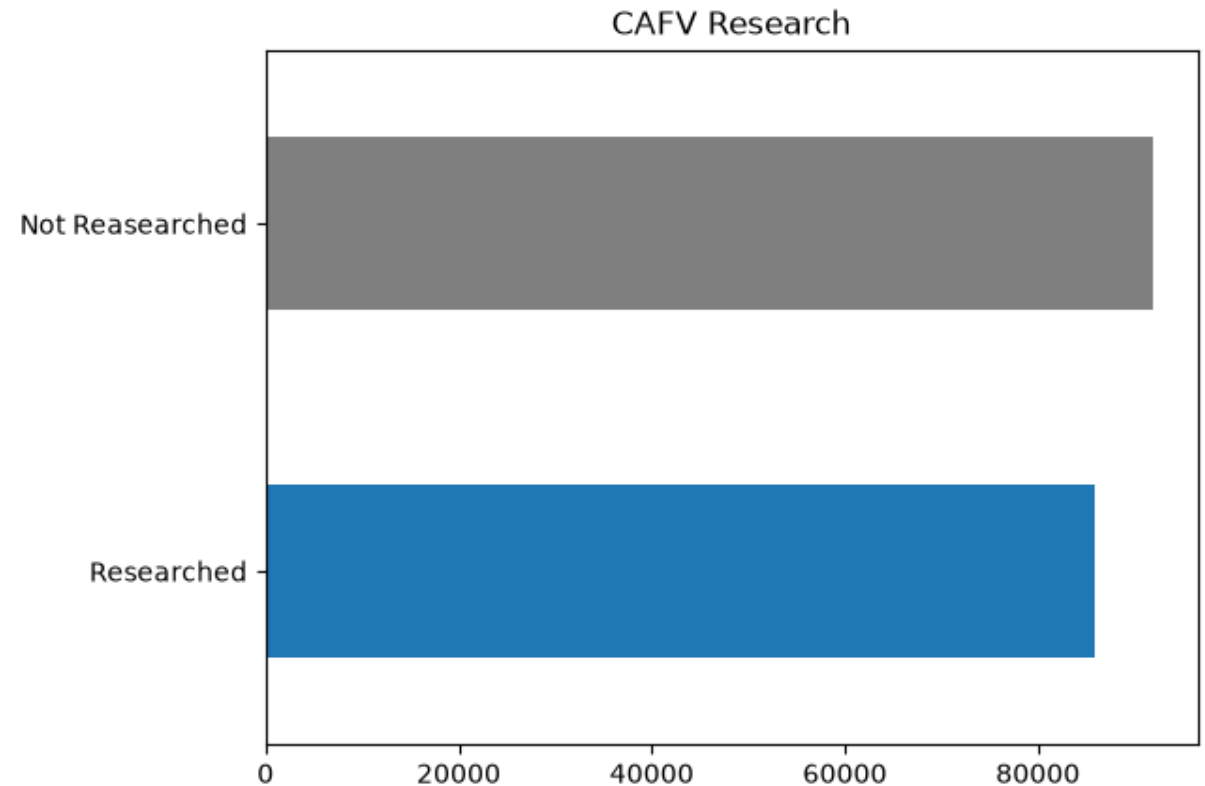
CAFV Eligibility

51.70%

Not researched for the CAFV

48.30%

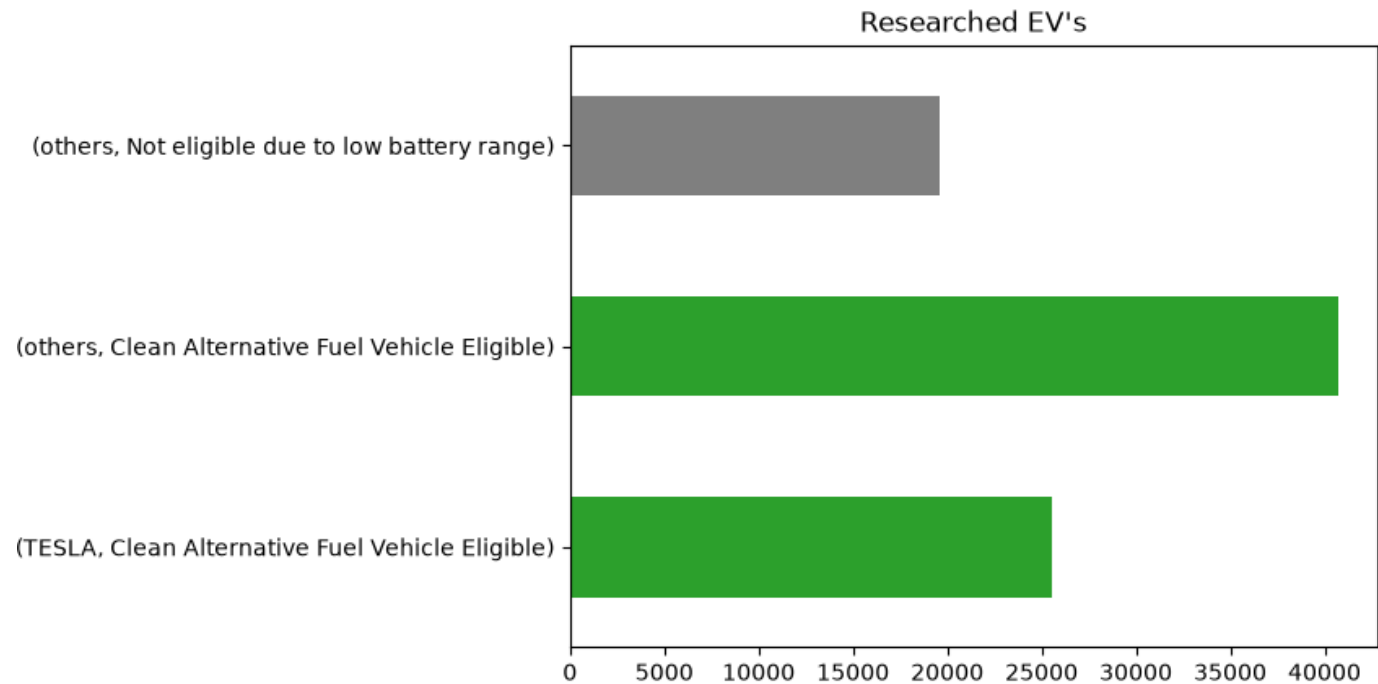
Researched for the CAFV



CAFV Eligibility

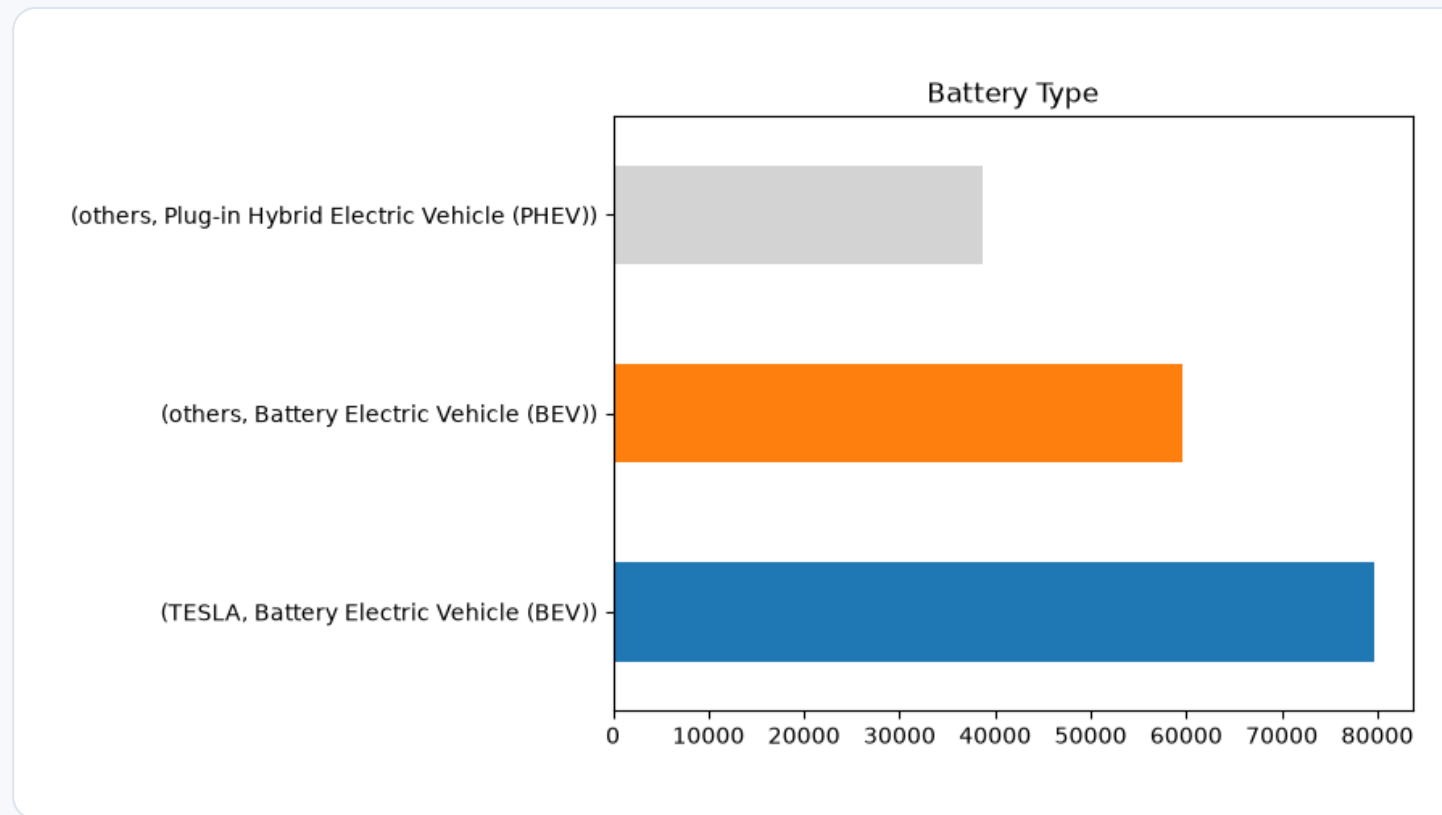
67.54%
Other vehicles

100%
TESLA vehicles



Electric Range

To Compare the Electric Range fairly we will exclude the Hybrid vehicles.



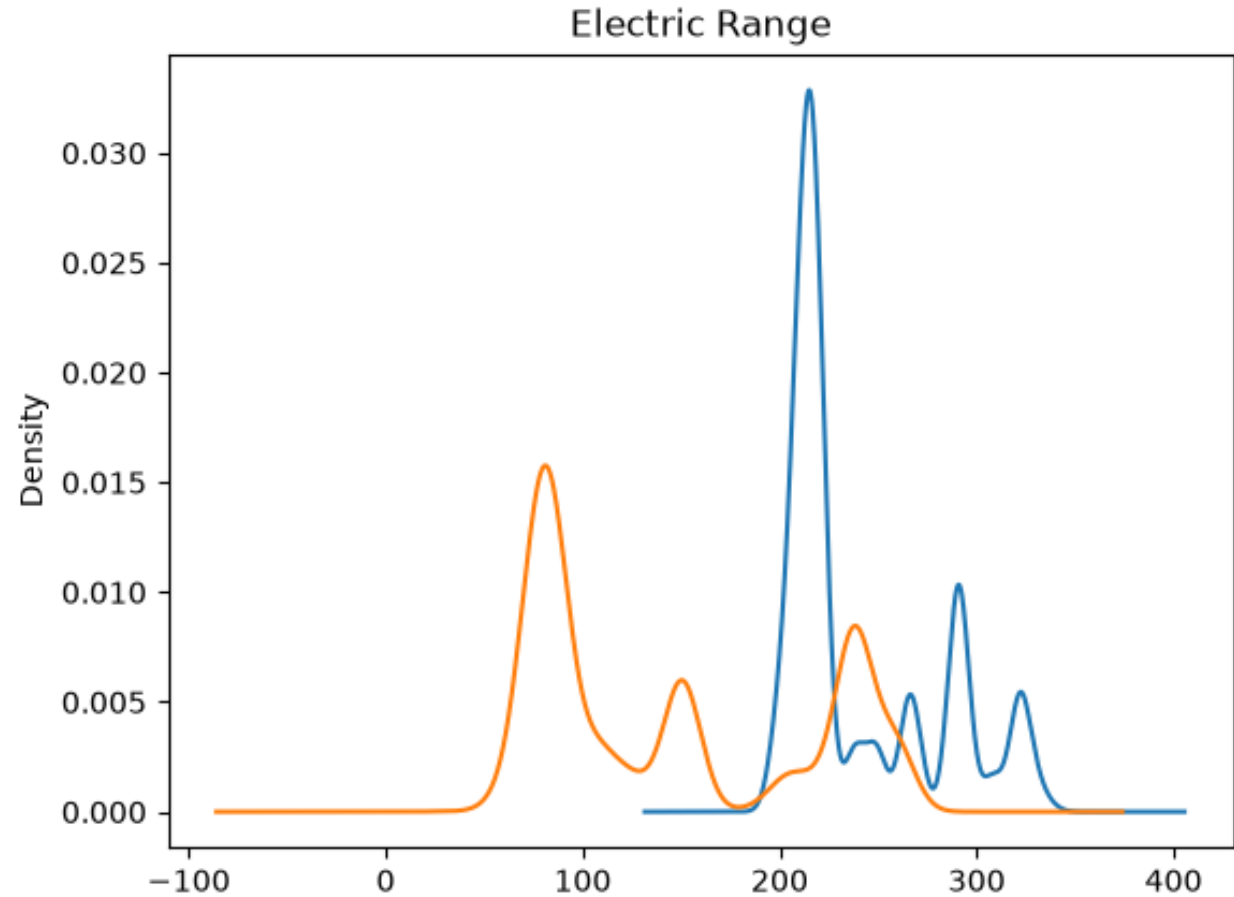
Electric Range

143.67

Others average range

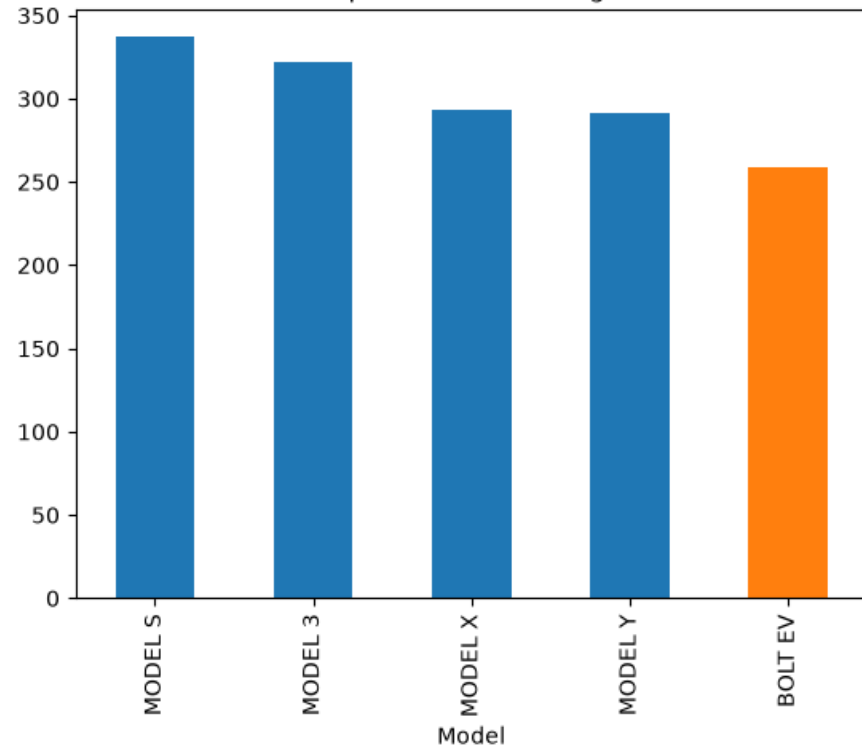
240.82

TESLA's average range

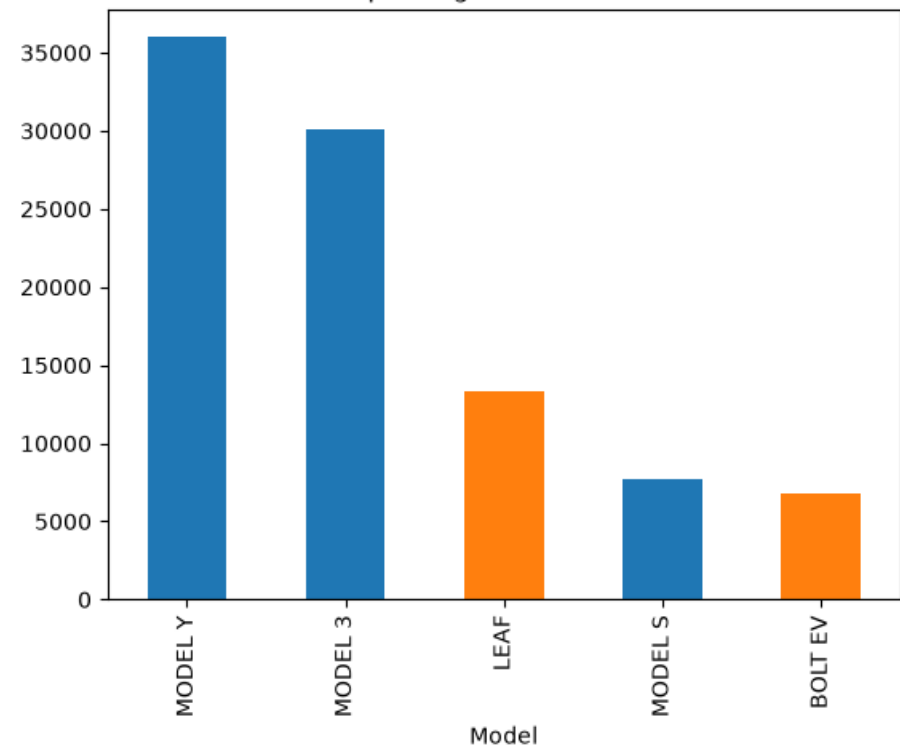


Top 5

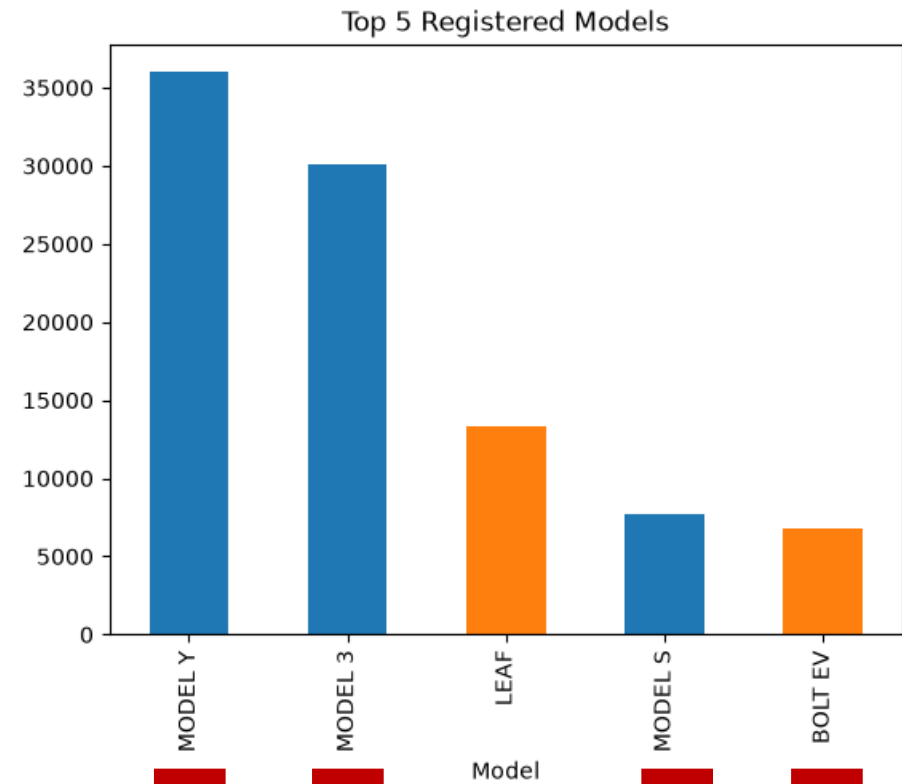
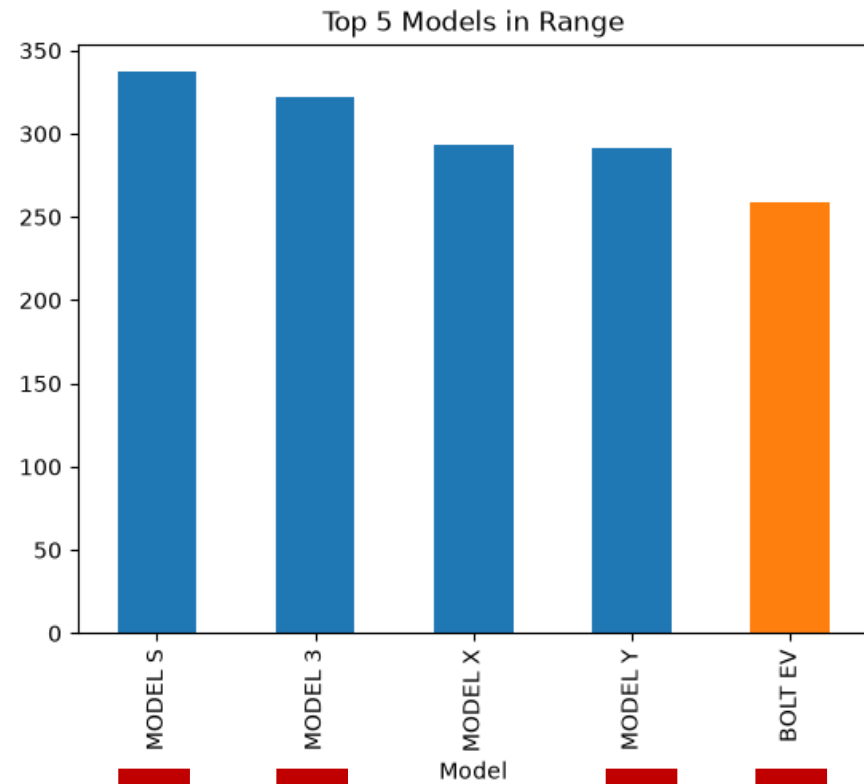
Top 5 Models in Range



Top 5 Registered Models



Top 5



4 of the 5 Models overlap across both lists

Conclusion

- The significant amount of missing Base MSRP data limited our ability to include vehicle price in the analysis
- Electric range appears to have a significant impact on the number of registered vehicles , suggesting that increasing the range of future models could help improve sales performance.
- All researched Tesla vehicles in the analysis were CAFV eligible, which may have contributed to their appeal while the tax incentive was available. However, new vehicle sales and leases after July 31, 2025 no longer qualify for this exemption.